MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION March 2026
Fourth SEMESTER
Discipline Specific Elective (DSE) - MG4DSECHEA03 - MANUFACTURE OF PETROCHEMICALS I
(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse (An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Answer any **5** questions

Each question carries **4** marks

1. State major engineering problems in Freush process. [K]/[CO1]

- Large amounts of heat are needed to generate superheated water (around 160–170°C) to melt underground sulfur, making the process energy-intensive and costly.
- Drilling and maintaining concentric pipes (three-pipe system) deep underground is difficult, and corrosion or mechanical failure can occur due to high temperature and pressure.
- Significant heat is lost to surrounding rock formations, reducing efficiency and requiring continuous heating to keep sulfur molten.
- Proper control of compressed air and molten sulfur flow is challenging; fluctuations can reduce recovery efficiency or cause operational hazards.
- The process is only suitable for sulfur deposits located under impermeable layers and at certain depths, restricting its use.
- Possible groundwater contamination and land subsidence can occur if the operation is not properly managed. (Any 4 Points give 4marks)

2. Discuss the role of monoethanol amine (MEA) in hydrogen purification. [U]/ [CO1]

- Monoethanol amine (MEA) is used to remove acidic gases such as carbon dioxide (CO_2) and hydrogen sulfide (H_2S) from crude hydrogen streams.
- MEA reacts chemically with CO_2 to form soluble carbamates, enabling effective separation from hydrogen gas.
- The process selectively absorbs impurities while allowing purified hydrogen (H_2) to pass through.
- The MEA solution can be regenerated by heating, which releases absorbed CO_2/H_2S and allows reuse of the solvent.
- Widely used in refineries and ammonia plants for producing high-purity hydrogen.
- Ensures hydrogen meets the required purity standards for industrial and chemical processes. (Any 4 Points give 4 marks)

3. What are branched polymers? Give examples. [K]/ [CO2]

Branched polymers are a type of polymer in which the main linear chain has one or more side chains (branches) attached to it. These branches arise due to secondary reactions during

polymerization, resulting in a non-linear structure. The presence of branching affects the physical properties of the polymer, such as lowering its density, crystallinity, and melting point compared to linear polymers, while often increasing its flexibility. Branched polymers are generally less tightly packed due to the irregular arrangement of chains. Common examples include low-density polyethylene (LDPE), glycogen, amylopectin. (Definition 3 marks Example 1 marks)

4. List the uses of plastics. [K]/ [CO3]

Plastics are widely used in various fields due to their lightweight, durability, corrosion resistance, and ease of molding. In daily life, they are extensively used for making household items such as containers, bottles, buckets, and furniture. In the packaging industry, plastics are used for bags, films, and food packaging materials to preserve and protect goods. In the electrical and electronics sector, plastics serve as insulating materials for wires, cables, switches, and appliances. They are also used in the construction industry for pipes, fittings, flooring, and insulation materials. In the medical field, plastics are used to manufacture syringes, gloves, IV bags, and other disposable equipment. Additionally, plastics are used in the automotive and aerospace industries to reduce weight and improve fuel efficiency. Thus, plastics play a vital role in modern life across multiple sectors. (4 marks for 4 points)

5. Discuss on reinforcements used in plastics and their advantages. [U]/ [CO3]

Reinforcements used in plastics include materials such as glass fibres, carbon fibres, asbestos, and synthetic fibres, which are incorporated into polymer matrices to improve their performance. During processing, these reinforcements play an important role by increasing the strength of the material without causing a significant increase in weight, thus maintaining a high strength-to-weight ratio. In the final product, they enhance key mechanical properties such as tensile strength, stiffness, and impact resistance, making the material more durable and reliable. The advantages of using reinforcements in plastics include improved mechanical strength, better dimensional stability, increased resistance to wear and deformation, and suitability for demanding applications in fields like automotive, aerospace, and construction. (Explanation 2 marks Advantages 2 marks)

6. List any four properties of elastomers. [K]/ [CO4]

Elastomers are polymers that exhibit high elasticity and flexibility due to their unique molecular structure. They can undergo large reversible deformations, meaning they can be stretched significantly and return to their original shape when the stress is removed. Elastomers generally have low Young's modulus and high elongation at break, indicating softness and extensibility. They possess good resilience and impact resistance, along with excellent resistance to abrasion and wear. Additionally, elastomers show good chemical resistance and can withstand a wide range of temperatures, making them suitable for applications such as rubber products, seals, and gaskets. (4 marks for 4 Points)

7. Describe nitrile rubber. [U]/ [CO4]

Nitrile rubber, also known as Buna-N (NBR), is a synthetic rubber obtained by the copolymerization of acrylonitrile and butadiene. The presence of polar nitrile groups (–CN) imparts excellent resistance to oils, fuels, and many chemicals, making it highly suitable for industrial applications. It possesses good mechanical strength, abrasion resistance, and moderate heat resistance, although its flexibility decreases at low temperatures. Nitrile rubber is widely used in the manufacture of oil-resistant products such as gaskets, seals, hoses, and fuel system components in automobiles and machinery due to its durability and chemical resistance. (Definition with 3 points 4 marks)

[4x5= 20]

Part B

Short Essay Questions

Answer any **5** questions

Each question carries **7** marks

8. Outline emulsion polymerization with its components. [An]/ [CO4]

Emulsion polymerization is a type of free radical polymerization carried out in an aqueous medium where the monomer is emulsified in water with the help of surfactants. In this process, the monomer forms small droplets stabilized by an emulsifying agent, and polymerization occurs mainly within micelles formed by the surfactant. A water-soluble initiator, such as potassium persulfate, decomposes to produce free radicals, which enter the micelles and initiate polymerization, leading to the formation of polymer particles dispersed in water (latex).

The main components of emulsion polymerization include: (1) Monomer, which is the substance to be polymerized (e.g., styrene, butadiene); (2) Water, acting as the dispersion medium; (3) Emulsifier or surfactant, which stabilizes the monomer droplets and forms micelles; (4) Initiator, usually water-soluble (e.g., persulfates), which generates free radicals; and sometimes (5) Buffer or stabilizers, to maintain pH and stability of the system. The advantages of this method include better heat control, high polymerization rate, and production of high molecular weight polymers, making it widely used in the manufacture of synthetic rubber, paints, and adhesives.

(Method- 4 marks components – 3marks)

9. Explain the manufacture of nitro benzene with reaction. [U]/[CO1]

Nitrobenzene is manufactured by the nitration of benzene using a nitrating mixture of concentrated nitric acid and concentrated sulfuric acid. In this process, benzene is treated with the acid mixture at about 50–60°C. Sulfuric acid acts as a catalyst and dehydrating agent, generating the active nitrating species, the nitronium ion (NO_2^+), from nitric acid. This nitronium ion then attacks the benzene ring in an electrophilic substitution reaction, replacing one hydrogen atom to form nitrobenzene.

The reaction is represented as: $C_6H_6 + HNO_3 \rightarrow C_6H_5NO_2 + H_2O$ (in presence of conc. H_2SO_4)

After the reaction, the mixture is allowed to settle, and nitrobenzene is separated, washed, and purified. This method is widely used in industry for the large-scale production of nitrobenzene, which is an important intermediate in the manufacture of aniline, dyes, and pharmaceuticals.

(Method 5 marks equation 2 marks)

10. Explain the properties and applications of thermoplastics and thermosetting plastics. [U]/[CO3]

Thermoplastics are linear or slightly branched polymers that soften on heating and harden on cooling, and this process is reversible. They possess properties such as flexibility, low to moderate strength, ease of moulding, recyclability, and good chemical resistance. Common examples include polyethylene (PE), polypropylene (PP), and polyvinyl chloride (PVC). Thermoplastics are widely used in applications such as packaging materials, plastic containers, pipes, electrical insulation, and household items due to their ease of processing and reusability.

In contrast, thermosetting plastics are highly cross-linked polymers that undergo a permanent chemical change when heated and cannot be remelted or reshaped. They are hard, rigid, and brittle with high thermal stability, excellent mechanical strength, and good resistance to chemicals and deformation. Examples include bakelite, urea-formaldehyde, and epoxy resins. These plastics are used in applications requiring durability and heat resistance, such as electrical fittings, adhesives, laminates, coatings, and automotive parts. Thus, thermoplastics are preferred for recyclable and

flexible applications, whereas thermosetting plastics are suitable for high-strength and heat-resistant uses. (Any 3 points from each type with one example give full credit)

11. Explain the preparation of nylon-6,6 using hexamethylene diamine and adipic acid with a neat reaction. [U]/[CO2]

Nylon-6,6 is prepared by the condensation polymerization of hexamethylene diamine and adipic acid. Initially, equimolar amounts of hexamethylene diamine ($H_2N–(CH_2)_6–NH_2$) and adipic acid ($HOOC–(CH_2)_4–COOH$) are mixed. This is then heated at about 250–280°C under pressure, during which condensation occurs with the elimination of water molecules, leading to the formation of long-chain polyamide known as nylon-6,6.

The reaction can be represented as: $n\ H_2N–(CH_2)_6–NH_2 + n\ HOOC–(CH_2)_4–COOH \rightarrow [–NH–(CH_2)_6–NH–CO–(CH_2)_4–CO–]_n + 2n\ H_2O$

This process results in a linear polymer with repeating amide (–CONH–) linkages, which impart high strength and durability. Nylon-6,6 thus produced is widely used in fibres, textiles, ropes, tyres, and engineering plastics due to its excellent mechanical strength, abrasion resistance, and thermal stability. (Explanation 5 marks equation 2 marks)

12. Describe the advantages and applications of transfer moulding. [U]/[CO3]

Transfer moulding is a method used mainly for moulding thermosetting plastics, where the material is first heated in a chamber and then transferred into a closed mould cavity. This process offers several advantages, such as better surface finish and dimensional accuracy compared to compression moulding, as the material flows uniformly into the mould. It is suitable for moulding complex shapes and intricate designs, including inserts like metal components, as it ensures proper encapsulation. The process also reduces internal stresses, minimizes material wastage, and provides improved mechanical properties in the final product.

The applications of transfer moulding are mainly found in the electrical and electronics industries for manufacturing components such as switches, circuit breakers, connectors, and encapsulated coils. It is also used in the production of automotive parts, appliance components, and moulded items requiring high precision and good surface finish. Thus, transfer moulding is widely preferred for producing high-quality, detailed thermosetting plastic products.

(3 points from advantages and applications give 3.5 marks)

13.Using the properties of PAN, explain its suitability for fibers, filtration membranes, and protective materials. [U]/[CO2]

Polyacrylonitrile (PAN) is a synthetic polymer characterized by the presence of strong polar nitrile (–CN) groups along its backbone, which impart several desirable properties. PAN exhibits high tensile strength, good thermal stability, excellent chemical resistance, and resistance to sunlight and weathering. It also has good fibre-forming ability due to strong intermolecular interactions between nitrile groups, resulting in closely packed, crystalline structures.

These properties make PAN highly suitable for fibres, where it is used to produce acrylic fibres that are lightweight, strong, and resistant to abrasion, with good resistance to chemicals and environmental degradation. In filtration membranes, PAN's chemical resistance and stability, along with its ability to form fine porous structures, make it effective for separating particles and impurities in water and air filtration systems. For protective materials, PAN is valuable due to its thermal stability and resistance to chemicals, and it is also used as a precursor for carbon fibres, which are widely used in protective clothing, aerospace components, and high-strength materials. Thus, the combination of strength, stability, and chemical resistance makes PAN suitable for diverse advanced applications.

(PAN explanation 1 mark. 2 marks each for any 2 points for the suitability for fiber, membrane and protective materials.)

14. Describe preparation, properties, and uses of nitroethane, 1-nitropropane and 2-nitropropane. [U]/ [CO1]

Nitroalkanes such as nitroethane, 1-nitropropane, and 2-nitropropane are prepared by treating alkyl halides with silver nitrite ($AgNO_2$). $C_2H_5X + AgNO_2 \rightarrow C_2H_5NO_2 + AgX$ (nitroethane) $C_3H_7X + AgNO_2 \rightarrow C_3H_7NO_2 + AgX$ (nitropropanes, giving both 1- and 2-isomers).

These compounds exhibit characteristic properties due to the presence of the electron-withdrawing $–NO_2$ group. They are polar, have relatively high boiling points, and are moderately soluble in water but highly soluble in organic solvents. Primary nitroalkanes (nitroethane and 1-nitropropane) show tautomerism (nitro–aci form) and are acidic in nature, whereas secondary nitroalkanes (2-nitropropane) are less acidic. They undergo reactions such as reduction to amines and condensation reactions.

In terms of uses, nitroethane is used as a solvent, fuel additive, and intermediate in organic synthesis. 1-Nitropropane is used as a solvent for paints, coatings, and inks, while 2-nitropropane is widely used as an industrial solvent and in the manufacture of pharmaceuticals, pesticides, and explosives. Thus, these nitroalkanes are important both as solvents and as intermediates in chemical industries. (Preparation 2 marks properties 2.5 marks and uses 2.5 marks)

[7x5 = 35]

Part C

Essay Questions

Answer any **1** question

Each question carries **15** marks

15. Describe the manufacture of polystyrene by bulk polymerisation. Explain the preparation steps, properties, and applications of polystyrene. [A]/ [CO2]

Polystyrene is manufactured by bulk polymerization (mass polymerization) using styrene monomer. In this method, pure styrene is taken in a reactor along with a small amount of initiator such as benzoyl peroxide. The mixture is heated to about 80–150°C in the absence of solvent. The initiator decomposes to form free radicals, which initiate chain polymerization of styrene. As the reaction proceeds, viscosity increases and the polymer is formed. The process is carried out in stages to control heat and avoid auto-acceleration. Finally, the molten polymer is cooled and solidified to obtain polystyrene.

The reaction can be represented as: $n\ CH_2=CH–C_6H_5 \rightarrow [–CH_2–CH(C_6H_5)–]_n$

Preparation steps: Initially, purified styrene monomer is mixed with an initiator. The mixture is heated gradually in a reactor under controlled temperature and pressure. Polymerization begins with free radical formation and continues with chain propagation. The increasing viscosity is managed by stepwise heating. After completion, the polymer is cooled, extruded, and cut into granules for commercial use.

Properties: Polystyrene is a transparent, rigid, and lightweight thermoplastic with good dimensional stability. It has excellent electrical insulating properties and is resistant to acids and alkalis but is attacked by organic solvents. It is brittle in nature and has low impact strength, though it can be modified (as high-impact polystyrene). It softens on heating and can be easily moulded.

Applications: Polystyrene is widely used in the manufacture of disposable cups, plates, and packaging materials (thermocol/expanded polystyrene). It is used for electrical insulation, toys, household items, laboratory ware, and as foam for thermal insulation in refrigerators and buildings.

Thus, due to its ease of processing, low cost, and useful properties, polystyrene has extensive commercial applications. (2.5 marks each for bulk polymerisation and preparation step 5 marks each for properties and applications)

16. Illustrate fibres and explain fibre spinning techniques. [A]/ [CO4]

Fibres are long, thin, hair-like polymeric materials characterized by a high length-to-diameter ratio, flexibility, and sufficient tensile strength, which enable them to be spun into yarns and fabrics. Fibres may be natural (cotton, wool, silk) or synthetic (nylon, polyester, acrylic). For a polymer to be suitable as a fibre, it should have linear structure, high molecular weight, strong intermolecular forces, and the ability to orient along the fibre axis to provide strength and durability.

Fibre spinning is the process of converting polymer materials into continuous filaments. The main fibre spinning techniques are:

1. Melt Spinning: In this method, the polymer is melted and extruded through a spinneret (a plate with fine holes). The molten filaments are cooled and solidified by air or water and then wound onto spools. It is simple, economical, and widely used for thermoplastic polymers like nylon, polyester, and polyethylene.
2. Dry Spinning: Here, the polymer is dissolved in a volatile solvent to form a solution, which is then extruded through a spinneret into a chamber of hot air. The solvent evaporates, leaving behind solid fibres. This method is used for polymers like cellulose acetate and acrylic fibres.
3. Wet Spinning: In this technique, the polymer solution is extruded into a coagulation bath containing a non-solvent, where the polymer precipitates as fibres. It is used for polymers that cannot be melted, such as rayon and some acrylic fibres.
4. Gel Spinning: This is a specialized method where the polymer is in a semi-solid (gel) state and extruded to form highly oriented fibres with very high strength and modulus. It is used for high-performance fibres like ultra-high molecular weight polyethylene.

Thus, fibre spinning techniques are essential for producing fibres with desired mechanical properties, and the choice of method depends on the nature of the polymer and its thermal and chemical stability. (Fiber explanation and example 5 marks different method 10 marks).

[15x1 = 15]